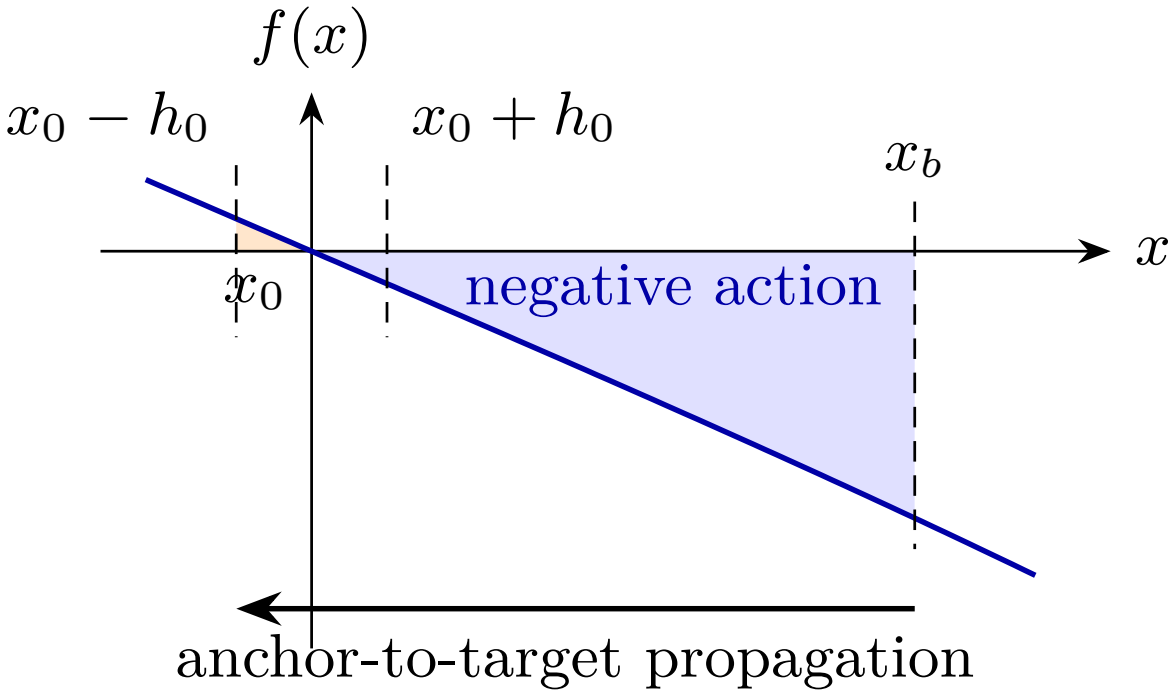
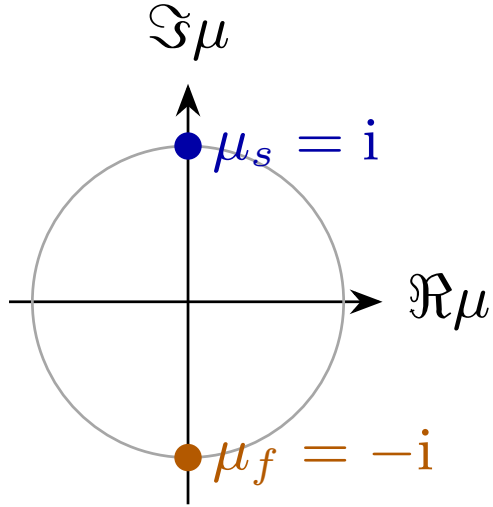




distinct roots
at $x = x_0,\ t = 0$